

32in Gravel Bike Spring 2026 – Cal Poly Bike Builders



Designed and manufactured a custom 32-inch-wheel gravel bike frame from concept through final assembly. Developed frame geometry through reverse engineering other bikes, fabricated from 0.035 in chromoly tubing, and TIG welded the frame using frame building fixtures. Designed metal 3D-printed rear dropouts integrating a SRAM UDH derailleur interface and flat-mount brake, reducing machining requirements while improving drivetrain and tire clearance.

Skills: Tig welding, Metal 3d Printing, Powder Coating

Organizations: Cal Poly Bike Builders

Featured in: [Bike Rumor](#)



Figure 1. Photos from MADE Bike Show 2026 (Portland, OR)

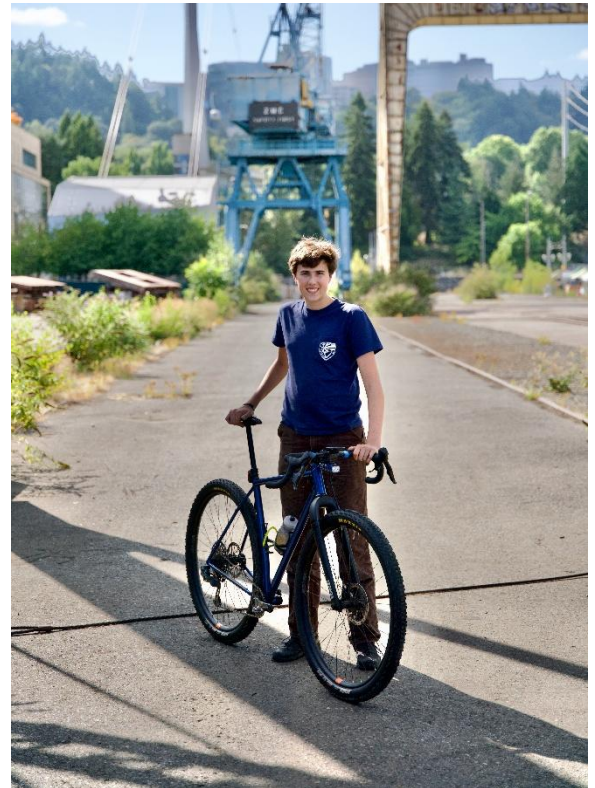


Figure 2. MADE Bike Show 2026



Figure 3. Welded front triangle



Figure 4. Front triangle in frame jig ready for welding



Figure 5. Tube in milling machine for mitering



Figure 6. Metal 3d printed dropouts and yoke



Figure 7. Frame prepped for powder coating



Figure 8. Powdered frame



Figure 9. Frame after baking

FPV Flying Wing Winter 2026

Designed, fabricated, and flight-tested a 3D-printed FPV flying wing with an emphasis on durability, modularity, and serviceability. Adapted an existing airframe design for additive manufacturing to simplify replacement of structural components and electronics modules. Integrated and configured a Matek F405 Wing flight controller running iNav, completing system assembly and flight testing to validate the airframe and control system.

Skills: 3d Printing, FPV, INAV



Figure 1. First iteration printed in standard PLA



Figure 2. Final version printed in LW-PLA being chased by an FPV Quadcopter

PCC Streetcar Electric Motor Upgrade

Summer 2026 – Lake Oswego, OR

Contributed to the modernization of a historic PCC streetcar by replacing original traction motors and resistor-based speed controls with modern brushless motors and electronics. Designed and fabricated custom motor mounts, electrical boxes and conduit, integrating the new drivetrain within the streetcar's existing mechanical constraints. Supported installation and integration of the propulsion hardware and high-voltage electrical system to enable operation without overhead power or a towed generator.

Skills: Welding, CAD

Organizations: Vintage Trolley, INC



Figure 1. New motor mounted in PCC truck with Retention Cage



Figure 2. Trolley 514 (already converted to battery) next to 513 which is being converted in the Lake Oswego Trolley Barn



Figure 3. Motor Inverter mounted on PCC truck next to Contactor Box

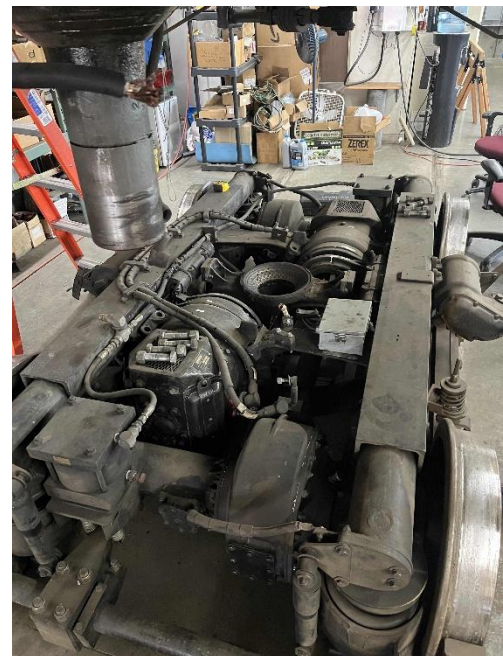
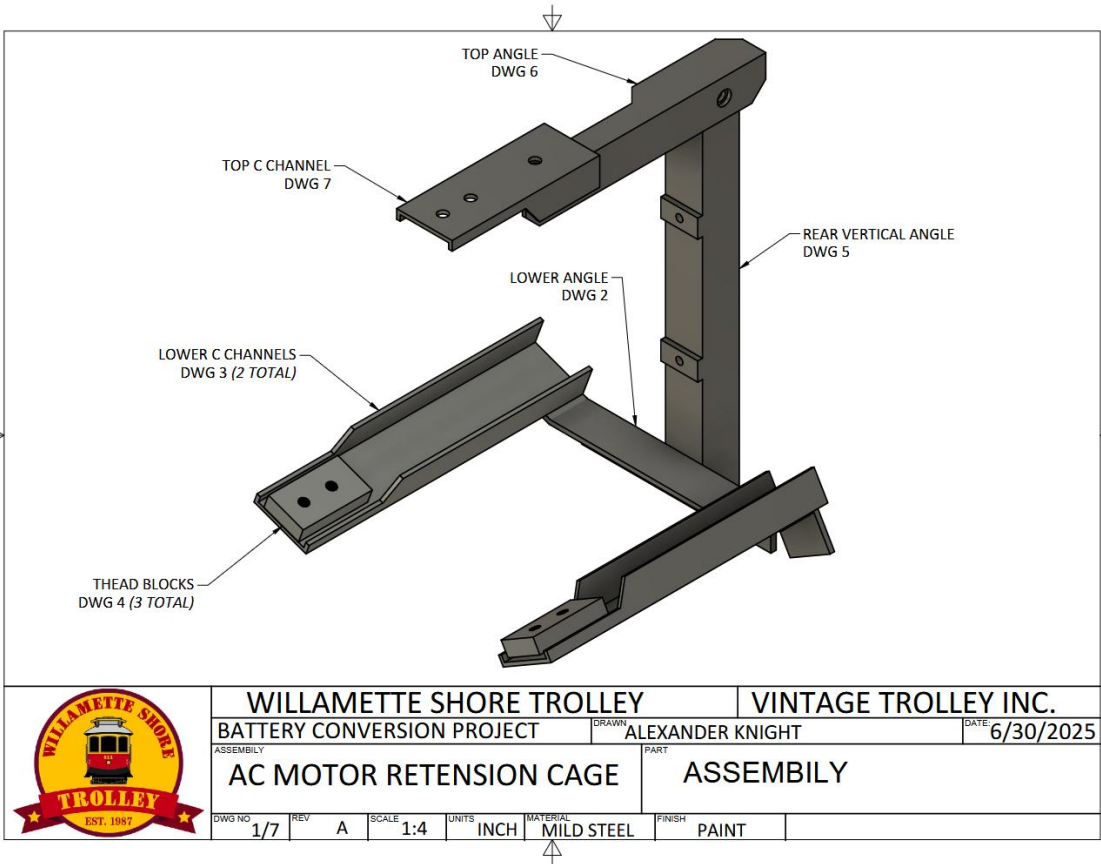
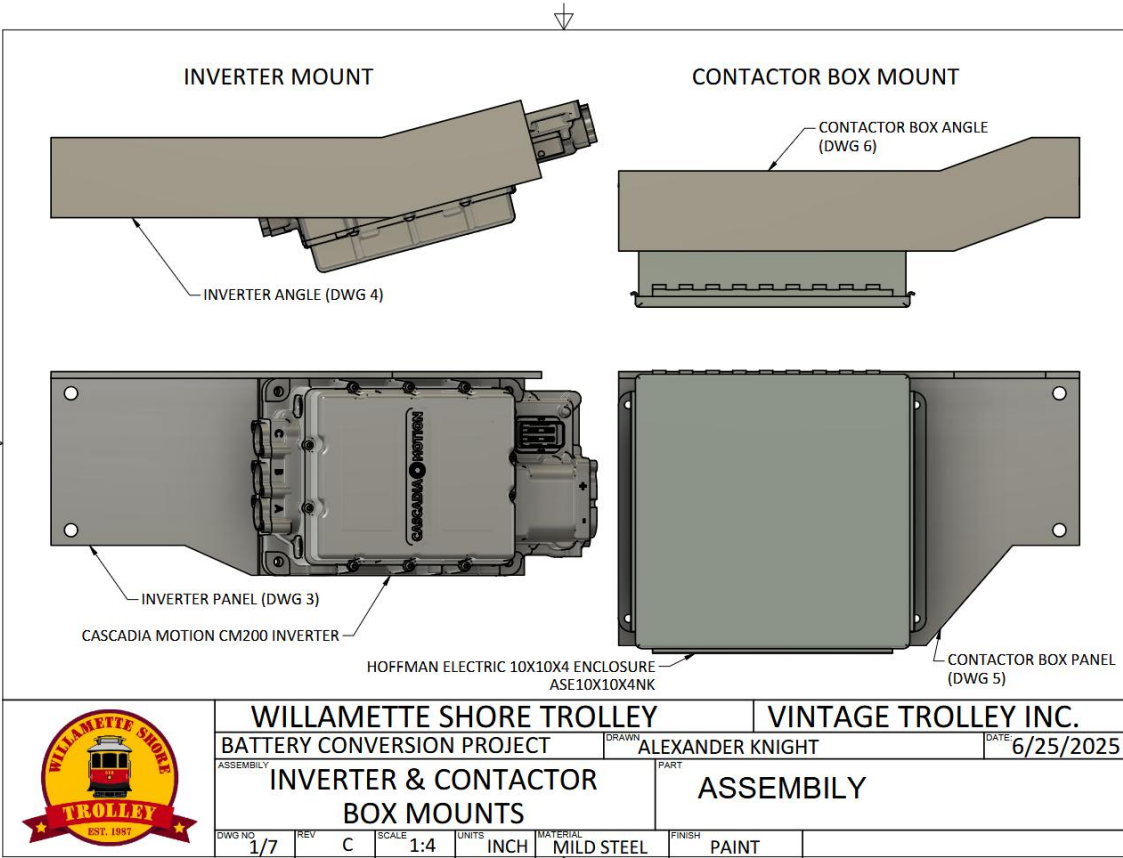


Figure 4. Original Truck configuration with traction motors



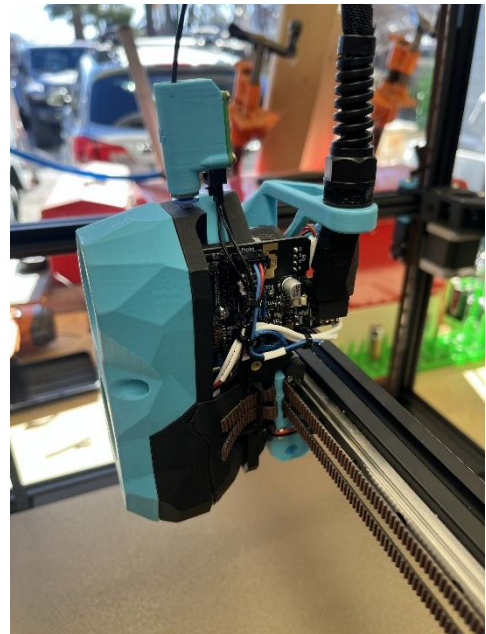
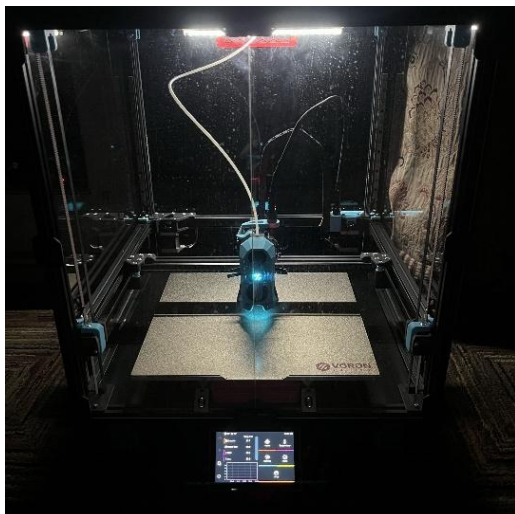
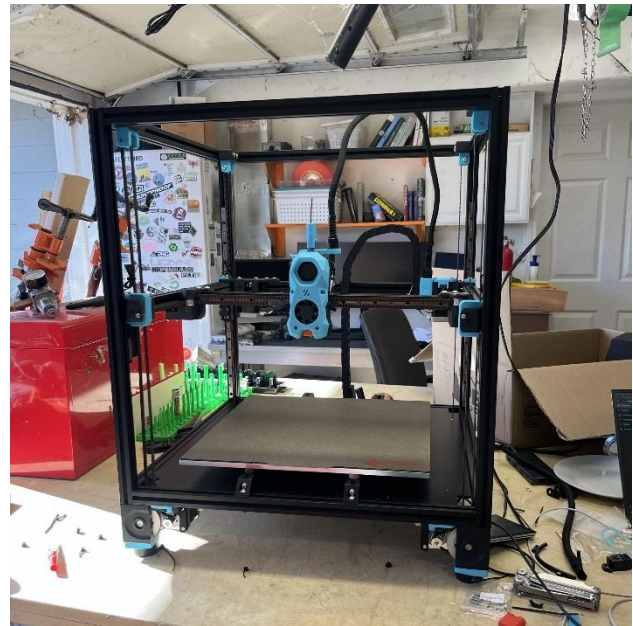
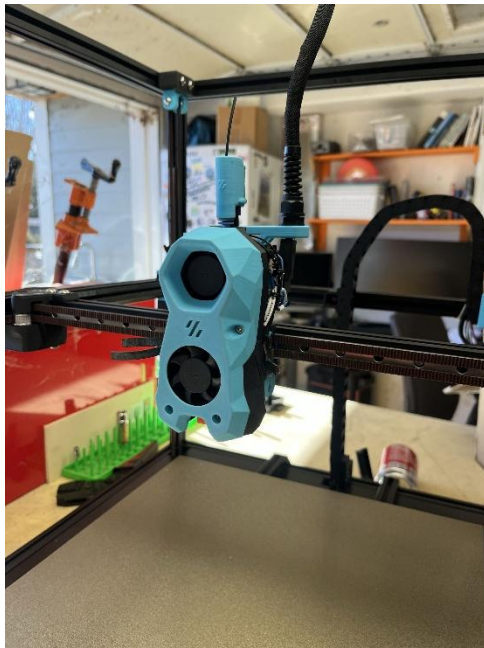


Voron 2.4 Winter 2025

Assembled, wired, configured, and calibrated a Voron 2.4 CoreXY 3D printer from individual components. Components we sourced online and printed at home and then integrated into a complete machine. Performed alignment and calibration to improve dimensional accuracy, reliability, and print quality. The project provided hands-on experience with precision machine assembly, motion control, mechatronics, and systematic troubleshooting.

Skills: 3D Printing, Linux, Soldering

Organizations: [Voron Design](#)



Vannon V2 Roof Rack Winter 2024

Redesigned Vannon Roof Rack to improve manufacturability, modularity, and shipping efficiency. Reworked the existing design to make it into two parts with only the rear half being van specific. This allows shipping in shorter boxes which reduces shipping costs.

Skills: CAD, CAM, Sheet Metal Design, CNC

Organizations: [Vannon](#)



Figure 1. Product Photo



Figure 2. Support leg designed to fit a socket wrench from the outside for easy installation



Figure 3. Prototype test fit on a 170 Sprinter Van

Vannon V2 Running Rail Winter 2024

Redesigned a running rail system for Mercedes Sprinter vans to improve manufacturability, modularity, and shipping efficiency. Reworked the existing design so components could be manufactured in-house and shipped in multiple sections, reducing packaging size and overall shipping cost. The modular architecture also reduced the number of unique SKUs while allowing customers to configure the rail system to match their specific van layout and installation requirements. Included outdoor Air and Water fittings with box to hold hoses

Skills: CAD, CAM, Sheet Metal Design, CNC

Organizations: [Vannon](#)



Figure 1. Product Photo

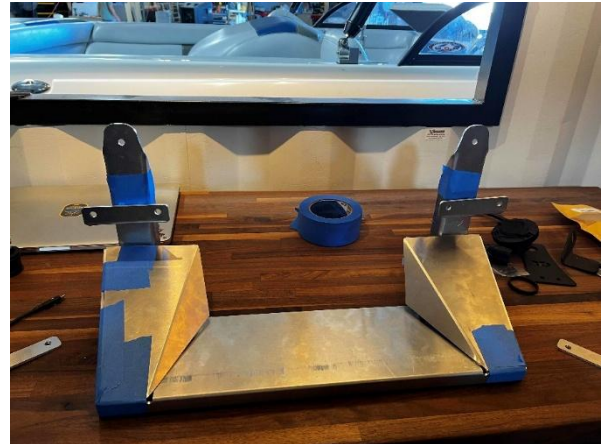


Figure 2. First Iteration of driver sidestep



Figure 3. Parts CNC cut for full van set



Figure 4. Bent Parts for passenger side

Vertical CNC Milling Machine Spring 2023

Designed and built a vertical CNC milling machine in high school metal shop. Drew the plans and built it using CNC milling, plasma cutting, water jet. Wired with NEMA 32 stepper motors and controlled using Linux CNC on a parallel port breakout board.

Skills: CNC, CAD, CAM, Plasma Cutting, Water Jet, Welding Linux CNC, Electronics

Recognition: Best in show at California Mid State Fair for Projects Built in a High School Shop

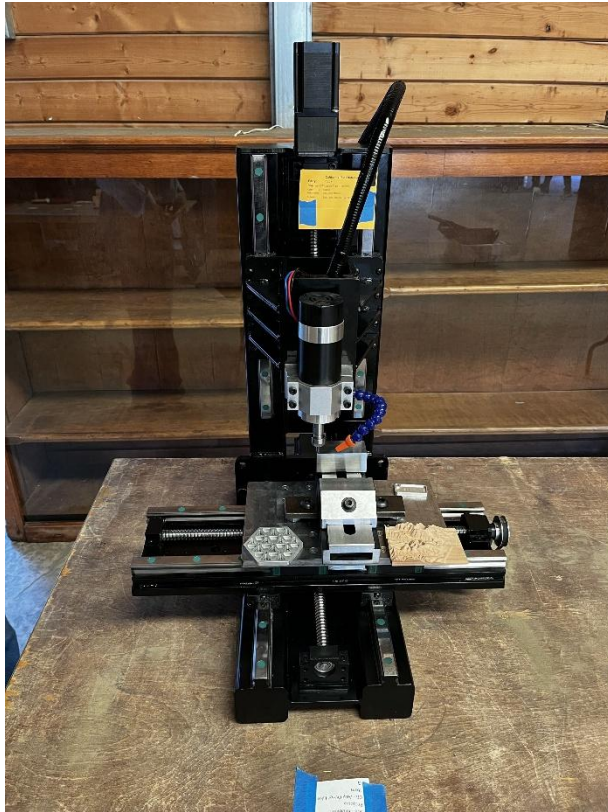


Figure 1. Finished Machine



Figure 2. Best in Show at California Mid State Fair



Figure 3. Spindle Spacer Test Fit



Figure 4. Calibrating Linear Rails



Figure 5. Test assembled before powder coat

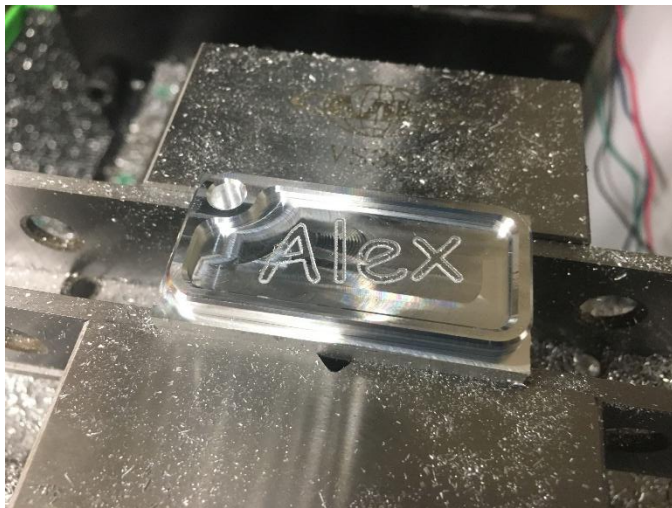


Figure 6. Test part in Aluminum



Figure 6. Column machined parallel in Haas CNC